
CSE 579 Course Project Report

Parametric Memory using LoRA

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Abstract

This project explores how LoRA can be used as a form of parametric memory to represent knowledge in large language models. It compares this approach with a retrieval-based method, RAG. RAG relies on retrieving external information during inference, which can be useful, but it may also make responses slower and less consistent. In contrast, LoRA stores domain-specific knowledge directly in the model parameters, helping the model produce faster and more stable answers.

The system was evaluated using a manually created cybersecurity question-answer dataset. The LoRA-based model achieved 100% accuracy on the test questions from the dataset, while the RAG-based system achieved about 53% accuracy. Although RAG sometimes retrieved useful information, its generated answers were often inconsistent or incorrect. The LoRA model, however, produced answers that were more accurate, stable, and closely matched the expected responses.

Overall, the results suggest that LoRA-based parametric memory is a more reliable and efficient approach for representing domain-specific knowledge.

1 Problem and Motivation

Large language models are commonly used in applications that require external knowledge. One popular method for supporting this is retrieval-augmented generation (RAG), where the model retrieves relevant information during inference before generating an answer. Although RAG can improve response quality, it also has some drawbacks. The retrieval process can increase response time, and the model may not always use the retrieved information correctly. This can lead to answers that are inconsistent or inaccurate.

These limitations motivate the need for a more efficient approach to knowledge representation. Instead of depending on external retrieval, this project studies parametric memory, where knowledge is stored directly in the model parameters. This allows the model to generate faster and more consistent responses without requiring an additional retrieval step.

In this project, LoRA is used as a parameter-efficient fine-tuning technique to add domain-specific knowledge to the model. This makes it possible to improve the model's knowledge while keeping the system lightweight and efficient.

2 Method / System Design

In this project, we explored two methods for knowledge representation in large language models: a retrieval-based approach using RAG and a parametric memory approach using LoRA. The main goal is to compare these methods based on accuracy, stability, and efficiency.

In the RAG approach, the cybersecurity dataset is first converted into vector embeddings using a sentence-transformer model. These embeddings are stored in a FAISS index, which allows the system

to search for similar information efficiently. During inference, the system retrieves the most relevant context for the input question and passes it to a GPT-2 model to generate the final answer. Although this method gives the model access to external knowledge, it adds extra steps and can sometimes produce inconsistent responses.

The LoRA-based approach works differently by storing knowledge directly inside the model parameters through lightweight fine-tuning. This reduces the need for retrieval during inference and allows the model to generate answers more efficiently and consistently. Because of this, LoRA provides a simpler and more stable way to handle domain-specific knowledge.

Both approaches were evaluated using the same cybersecurity question-answer dataset and the same test questions to ensure a fair comparison. From a Knowledge Representation and Reasoning (KRR) perspective, the main difference is how knowledge is represented: RAG uses external retrieved text, while LoRA stores knowledge internally in the model parameters.

3 Experimental Setup

The experiments were conducted to compare two approaches: a retrieval-based system using RAG and a parametric memory model using LoRA. The evaluation focuses on three main factors: accuracy, stability, and efficiency.

For the RAG system, the cybersecurity dataset is first converted into vector embeddings using a sentence-transformer model. These embeddings are stored in a FAISS index, which allows the system to search for relevant information efficiently. During inference, the system retrieves the top- k most relevant entries for each question and passes the retrieved context to a GPT-2 model to generate the final answer. Although this gives the model access to external knowledge, it also adds extra processing during inference.

The LoRA-based model uses a different approach. GPT-2 is fine-tuned with LoRA by updating only a small number of parameters, allowing cybersecurity knowledge to be stored directly inside the model. This removes the need for retrieval during inference and helps the model generate responses more efficiently and consistently.

To make the comparison fair, both models were tested using the same cybersecurity question-answer dataset and the same fixed set of test questions.

4 Results and Analysis

The LoRA-based model showed strong accuracy and consistency because it stores knowledge directly in the model parameters. When tested on 15 questions from the dataset, the model answered all of them correctly, achieving 100% accuracy. This suggests that the model successfully learned and retained the cybersecurity knowledge. The training loss also decreased to about 0.13, which indicates stable training and good convergence.

In comparison, the RAG-based system achieved about 53% accuracy on the same 15 test questions. Although it retrieved useful information in some cases, some of its answers were incorrect or inconsistent. This happened mainly because the system depends on external retrieval and may not always combine the retrieved context with the generation process effectively.

Overall, the results show that the LoRA-based model produced more accurate and stable responses. The RAG approach, however, showed a trade-off between consistency and variability depending on the decoding settings.

5 Conclusion

This project examined the use of LoRA-based parametric memory for knowledge representation in large language models, with RAG used as the comparison baseline. The results showed that the LoRA-based model performed better in terms of accuracy, stability, and consistency because it stored knowledge directly in the model parameters. This allowed the model to generate reliable answers without depending on external retrieval.

In contrast, the RAG-based approach produced more variable results because it relied on retrieved context and generation settings. Overall, the findings suggest that LoRA-based parametric memory is a reliable and efficient approach for domain-specific knowledge tasks, especially when accuracy and consistency are important.

The code and resources for this project are available at: [Project Repository](#)

Contribution Table

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Member	Contribution	Share / Role
Wala Aljohani	analysis, presentation, editing system design, writing	30%
Jawaher Alzahrani	evaluation, analysis, writing , editing	35%
Razan Alshehri	data preparation, baseline implementation, experiments	35%

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